Lecture 4: Section 0.5

Exponential and Logarithmic Functions

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Exponential Functions

Definition

An exponential function has the form:

$$f(x) = a^x \quad \text{where } a > 0 \text{ and } a \neq 1$$

Key Features

- Domain: $(-\infty, \infty)$
- Range: $(0, \infty)$
- y-intercept: $(0, 1)$
- Horizontal asymptote: $y = 0$
- $a > 1$: Increasing function
- $0 < a < 1$: Decreasing function

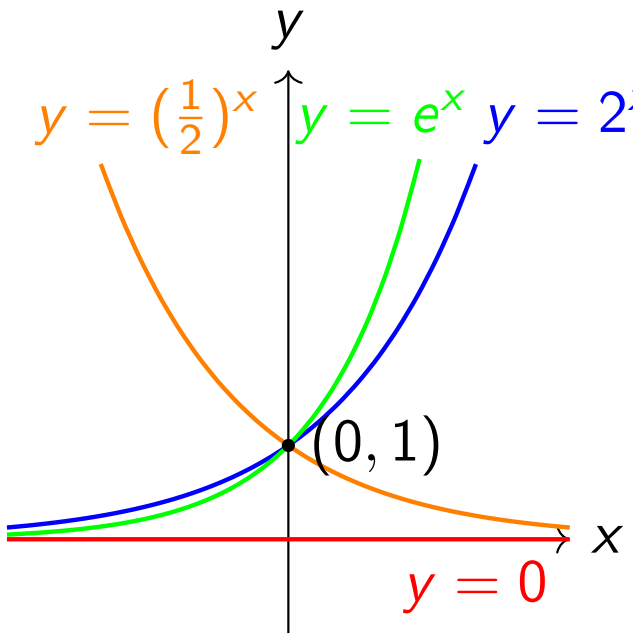
Exponential Functions

Example

Graph and analyze:

1. $f(x) = 2^x$
2. $f(x) = \left(\frac{1}{2}\right)^x$
3. $f(x) = e^x$
4. $f(x) = 3^{x-2} + 1$

Graph: Exponential Functions



Logarithmic Functions

Definition

A logarithmic function has the form:

$$f(x) = \log_a x \quad \text{where } a > 0, a \neq 1$$

It is the inverse of $f(x) = a^x$.

Key Features

- Domain: $(0, \infty)$
- Range: $(-\infty, \infty)$
- x-intercept: $(1, 0)$
- Vertical asymptote: $x = 0$
- $a > 1$: Increasing function
- $0 < a < 1$: Decreasing function

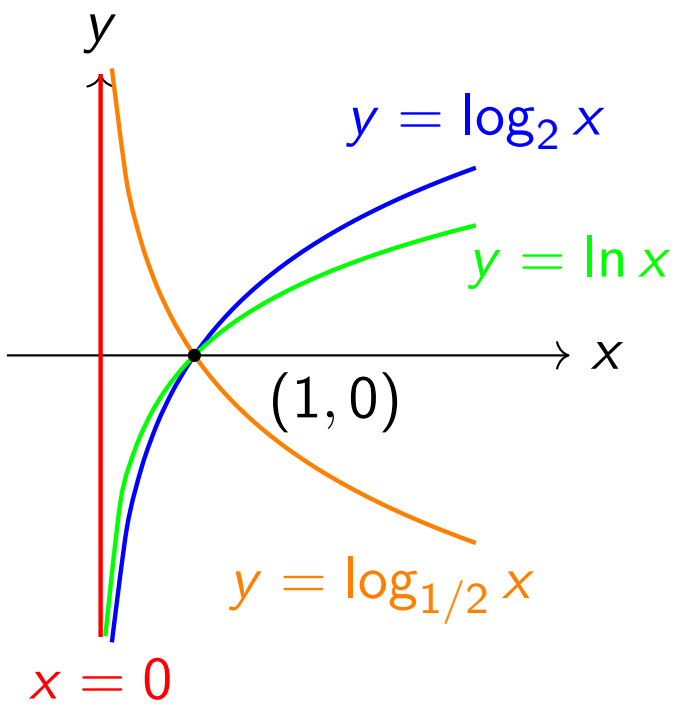
Logarithmic Functions

Example

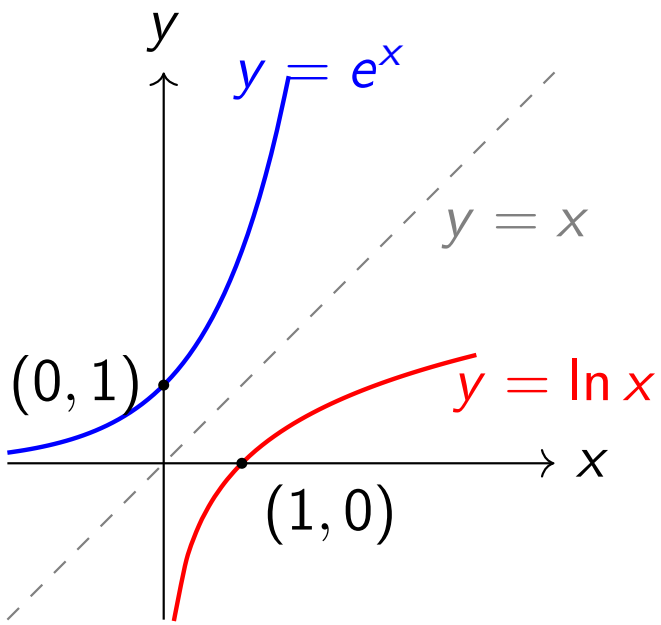
Graph and analyze:

1. $f(x) = \log_2 x$
2. $f(x) = \log_{1/2} x$
3. $f(x) = \ln x$
4. $f(x) = \log_3(x + 2)$

Graph: Logarithmic Functions



Graph: Exponential and Logarithmic Functions as Inverses



Inverse Relationship

- $e^{\ln x} = x$ for $x > 0$
- $\ln(e^x) = x$ for all real x
- Graphs are reflections across $y = x$

Properties of Exponential and Logarithmic Functions

For $a > 0$, $a \neq 1$:

- $a^m \cdot a^n = a^{m+n}$
- $\frac{a^m}{a^n} = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $a^0 = 1$
- $a^{-n} = \frac{1}{a^n}$
- $\log_a(xy) = \log_a x + \log_a y$
- $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$
- $\log_a(x^n) = n \log_a x$
- $\log_a 1 = 0$
- $\log_a a = 1$
- Change of base: $\log_a x = \frac{\ln x}{\ln a}$

Practice Problems: Exponential and Logarithmic

Graphing and Analysis

1. Graph $f(x) = 2^{x+1} - 3$ and find its domain, range, and asymptotes
2. Graph $f(x) = \log_3(x-2) + 1$ and find its domain, range, and intercepts
3. Find the inverse of $f(x) = 4e^{2x-1}$

Practice Problems: Exponential and Logarithmic

Properties and Equations

1. Solve: $2^{3x-1} = 16$
2. Solve: $\log_2(x+3) + \log_2(x-1) = 3$
3. Simplify: $\ln(e^3) + \log_4 16 - e^{\ln 5}$
4. Use properties to expand: $\ln\left(\frac{x^2\sqrt{y}}{z^3}\right)$

Key Features and Summary

Exponential Functions

- Domain: $(-\infty, \infty)$, Range: $(0, \infty)$
- Always pass through $(0, 1)$
- Horizontal asymptote: $y = 0$
- Rapid growth/decay

Logarithmic Functions

- Domain: $(0, \infty)$, Range: $(-\infty, \infty)$
- Always pass through $(1, 0)$
- Vertical asymptote: $x = 0$
- Inverse of exponential functions

Key Features and Summary

Applications

- Population growth, radioactive decay
- Compound interest, sound intensity
- Earthquake magnitude, pH scale
- Angle measurement, navigation

Solving Exponential Equations: Same Base

Method 1: Same Base

If $a^u = a^v$, then $u = v$.

Example

Solve each exponential equation:

1. $2^{3x-1} = 16$
2. $3^{x^2} = 3^{5x-6}$
3. $5^{2x+1} = \dfrac{1}{125}$
4. $4^{x-3} = 8^{2x+1}$
5. $e^{2x-1} = e^{5x+3}$

Step-by-Step Solution for $2^{3x-1} = 16$

$$
\begin{aligned}
2^{3x-1} &= 16 \\
2^{3x-1} &= 2^4 \\
3x - 1 &= 4 \\
3x &= 5 \\
x &= \frac{5}{3}
\end{aligned}
$$

Solving Exponential Equations: Logarithms

Method 2: Using Logarithms

If $a^u = b$, then $u = \log_a b$ or $u = \dfrac{\log b}{\log a}$.

Example

Solve using logarithms:

1. $3^x = 7$
2. $5^{2x-1} = 12$
3. $2^{x+3} = 5^{x-1}$
4. $e^{3x} = 15$
5. $10^{2x} = 45$

Step-by-Step Solution for $3^x = 7$

$$
\begin{aligned}
3^x &= 7 \\
\log(3^x) &= \log 7 \\
x \log 3 &= \log 7 \\
x &= \frac{\log 7}{\log 3} \approx \frac{0.8451}{0.4771} \approx 1.771
\end{aligned}
$$

Solving Logarithmic Equations: Single Logarithm

Method 1: Single Logarithm

If $\log_a u = \log_a v$, then $u = v$ (with $u, v > 0$).

Example

Solve each logarithmic equation:

1. $\log_2(x+3) = \log_2(2x-1)$
2. $\ln(x^2-4) = \ln(3x)$
3. $\log_5(2x-1) = \log_5(x+4)$
4. $\log(x-2) + \log(x+3) = \log 14$
5. $\ln(x+1) - \ln(x-2) = \ln 4$

Step-by-Step Solution for $\log_2(x+3) = \log_2(2x-1)$

$$\log_2(x+3) = \log_2(2x-1)$$
$$x+3 = 2x-1$$
$$3+1 = 2x-x$$
$$4 = x$$

Check: $\log_2(4+3) = \log_2 7$ and $\log_2(8-1) = \log_2 7$

Solving Logarithmic Equations: Exponential Form

Method 2: Exponential Form

If $\log_a u = v$, then $u = a^v$.

Example

Solve using exponential form:

1. $\log_3 x = 4$
2. $\log_2(3x - 1) = 5$
3. $\ln(x + 2) = 3$
4. $\log(x - 1) = 2$
5. $\log_4(2x + 6) = \frac{3}{2}$

Step-by-Step Solution for $\log_3 x = 4$

$$\log_3 x = 4$$
$$x = 3^4$$
$$x = 81$$

Check: $\log_3 81 = 4$ since $3^4 = 81$

Solving Complex Logarithmic Equations

Method: Combine and Solve

1. Use logarithmic properties to combine terms
2. Convert to exponential form if needed
3. Solve the resulting equation
4. Check for extraneous solutions

Example

Solve these complex equations:

1. $\log_2(x+1) + \log_2(x-1) = 3$
2. $\log_3(x+2) - \log_3(x-1) = 1$
3. $2\log_4 x - \log_4(x-1) = 1$
4. $\log(x+3) - \log(x-2) = 1$
5. $\ln(x+4) + \ln(x-1) = \ln(3x)$

Step-by-Step: Complex Logarithmic Equation

Solving $\log_2(x+1) + \log_2(x-1) = 3$

$$\begin{aligned}
\log_2(x+1) + \log_2(x-1) &= 3 \\
\log_2[(x+1)(x-1)] &= 3 \\
\log_2(x^2-1) &= 3 \\
x^2 - 1 &= 2^3 \\
x^2 - 1 &= 8 \\
x^2 &= 9 \\
x &= \pm 3
\end{aligned}$$

Check domain: $x+1>0$ and $x-1>0 \quad x>1$

$x=3$: $\log_2 4 + \log_2 2 = 2+1=3$

$x=-3$: Invalid (domain restriction)

Final answer: $x=3$

Applications and Real-World Problems: Exponential Growth/Decay

- Population growth: $P(t) = P_0 e^{kt}$
- Radioactive decay: $A(t) = A_0 e^{-kt}$
- Compound interest: $A = P\left(1 + \frac{r}{n}\right)^{nt}$

1. A bacteria culture doubles every 3 hours. If initially there are 500 bacteria, when will there be 8000 bacteria?
2. An investment of \$1000 earns 5% annual interest compounded continuously. How long until it doubles?
3. Carbon-14 has a half-life of 5730 years. If a fossil has 25% of its original Carbon-14, how old is it?
4. The pH of a solution is given by $\text{pH} = -\log[H^+]$. If $\text{pH} = 4.2$, find $[H^+]$.
5. Sound intensity: $L = 10 \log\left(\frac{I}{I_0}\right)$. If $L = 85$ dB, find $\frac{I}{I_0}$.

Practice Problems: Basic Level

Exponential Equations

1. $4^x = 64$
2. $2^{3x-1} = 32$
3. $5^{x+2} = 125$
4. $3^{2x} = 81$
5. $e^{2x} = 7$

Logarithmic Equations

1. $\log_5 x = 3$
2. $\log_2(3x - 1) = 4$
3. $\ln(x + 1) = 2$
4. $\log_4(2x) = \frac{1}{2}$
5. $\log(x - 3) = 1$

Practice Problems: Intermediate Level

Mixed Exponential Equations

1. $2^{x+3} = 5^{x-1}$
2. $3^{2x} - 3^x - 6 = 0$ (Hint: Let $u = 3^x$)
3. $4^x + 2^{x+1} - 8 = 0$
4. $e^{2x} - 3e^x + 2 = 0$
5. $5^{x-1} = 3^{x+2}$

Mixed Logarithmic Equations

1. $\log_3(x+1) + \log_3(x-1) = 1$
2. $\log_2(x+3) - \log_2(x-1) = 2$
3. $2\log_5 x = \log_5(2x+3)$
4. $\ln(x+2) - \ln(x-1) = 1$
5. $\log(x) + \log(x-3) = 1$

Practice Problems: Advanced Level

Challenging Equations

1. $\log_2(x^2 - 1) - \log_2(x - 1) = 3$
2. $2^{2x} - 6 \cdot 2^x + 8 = 0$
3. $\log(x + 2) - \log(x - 1) = \log 4 - \log 3$
4. $4^{x+1} = 8^{x-2}$
5. $\ln(x^2 - 4) - \ln(x + 2) = \ln(3x - 5)$

Application Problems

1. How long does it take \$5000 to double at 4% annual interest compounded monthly?
2. A radioactive substance decays from 100g to 25g in 20 years. Find its half-life.
3. The population of a city grows from 50,000 to 75,000 in 10 years. When will it reach 100,000?
4. If the Richter scale measures $M = \log\left(\frac{A}{A_0}\right)$ and one earthquake is 100 times stronger than another, what's the difference in magnitudes?

Common Pitfalls and Tips

Common Mistakes to Avoid

- Forgetting domain restrictions for logarithms
- Not checking for extraneous solutions
- Misapplying logarithmic properties
- Confusing $\log_a(x + y)$ with $\log_a x + \log_a y$
- Forgetting that $\log_a 1 = 0$ and $\log_a a = 1$

Problem-Solving Strategies

- Always check the domain first
- Use substitution for complex exponential equations
- Combine logarithmic terms when possible
- Verify solutions by substitution
- Use change of base formula when needed

Key Formulas to Remember

- $a^{\log_a x} = x$
- $\log_a a^x = x$
- $\log_a(xy) = \log_a x + \log_a y$
- $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$
- $\log_a(x^n) = n \log_a x$